

TANGUY, Russian Federation

WMO: 304050

Lat: 55.383N

Lon: 101.033E

Elev: 1387

StdP: 13.97

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.3	-32.1	-41.3	0.5	-37.0	-36.0	0.7	-31.9	14.3	12.5	12.2	8.5	0.6	50	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	84.3	65.9	80.9	64.4	77.8	63.2	68.4	79.2	66.6	77.2	64.9	74.5	4.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	96.7	73.6	62.9	90.4	71.3	61.2	84.9	69.8	33.7	78.9	32.2	77.6	30.8	74.6	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	9.9	8.5	DB	-41.5	90.4	7.8	4.0	-47.1	93.3	-51.6	95.6	-56.0	97.8	-61.7	100.8	(4)
(5)			WB	-41.0	71.6	6.8	2.3	-45.8	73.3	-49.8	74.6	-53.6	75.9	-58.5	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.5	-6.0	0.2	18.1	35.2	47.5	61.3	65.8	60.7	47.3	33.2	13.6	-0.9		
	DBStd	27.00	14.31	14.16	12.24	8.37	8.43	7.45	5.14	6.52	7.47	9.02	13.69	15.32		
	HDD50	8067	1738	1394	990	447	153	7	0	6	140	522	1092	1578		
	HDD65	12383	2203	1814	1455	893	545	156	52	162	532	986	1542	2043		
	CDD50	1312	0	0	0	4	76	344	491	337	59	1	0	0		
	CDD65	154	0	0	0	0	2	44	78	29	1	0	0	0		
	CDH74	1584	0	0	0	0	61	552	649	314	8	0	0	0		
	CDH80	370	0	0	0	0	12	140	156	62	0	0	0	0		
	WSAvg	3.5	2.8	2.7	3.5	4.2	4.5	3.9	3.6	3.4	3.7	3.9	3.6	2.8		
	PrecAvg	14.8	0.6	0.4	0.4	0.6	1.3	2.4	2.4	2.7	1.4	0.9	0.9	0.7		
	PrecMax	21.2	0.8	0.9	0.7	2.0	2.4	5.6	4.2	5.3	3.2	2.0	1.8	1.8		
(15) Precipitation	PrecMin	8.2	0.2	0.2	0.1	0.1	0.5	0.7	0.4	0.8	0.2	0.2	0.3	0.4		
	PrecStd	3.8	0.2	0.2	0.2	0.4	0.6	1.3	1.1	1.2	0.7	0.5	0.4	0.4		
	DB	27.8	33.8	48.2	64.0	80.8	88.9	89.0	85.9	74.8	58.2	43.6	31.8			
	MCWB	26.2	28.2	37.6	47.1	57.8	66.0	67.7	66.9	60.7	47.5	37.2	28.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	22.6	29.1	42.8	58.6	73.8	84.2	84.4	81.2	69.1	53.0	38.1	26.4			
	2% MCWB	20.8	25.9	34.1	44.9	55.0	64.9	66.9	65.3	57.3	45.3	33.8	24.7			
	5% DB	18.1	24.5	38.6	53.6	68.4	80.6	80.7	77.4	63.8	48.9	34.6	22.9			
	5% MCWB	16.8	21.9	31.8	42.2	52.5	62.8	66.1	63.6	54.5	43.3	31.4	21.4			
	10% DB	13.5	19.8	35.1	49.3	63.4	76.8	77.6	73.4	59.5	45.5	31.6	19.7			
	10% MCWB	12.4	17.8	29.9	39.7	49.1	61.4	64.9	62.4	52.7	40.5	29.3	18.3			
	0.4% WB	26.7	30.0	38.5	49.2	58.9	70.5	71.7	69.3	62.1	49.7	38.1	28.9			
	0.4% MCDB	28.0	33.3	47.0	61.1	77.9	82.5	83.2	81.0	72.7	55.6	42.1	31.2			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	21.0	26.0	35.0	45.9	56.1	66.7	69.1	66.9	58.2	46.5	34.2	24.8			
	2% MCDB	22.5	29.1	41.6	55.8	70.5	79.5	79.4	78.3	66.3	51.5	37.1	26.4			
	5% WB	16.9	22.0	32.5	43.1	53.6	64.6	67.5	64.8	55.6	43.9	31.8	21.6			
	5% MCDB	18.2	24.7	38.2	52.8	66.3	76.9	77.3	74.7	62.7	48.7	34.4	22.9			
	10% WB	12.5	17.8	29.8	40.2	50.9	62.6	66.0	62.9	52.9	40.9	29.2	18.3			
	10% MCDB	13.5	19.7	34.8	48.9	61.8	73.5	75.5	71.5	58.5	44.6	31.7	19.7			
	MDBR	14.8	19.5	21.8	18.0	19.8	19.6	17.5	17.5	16.3	13.8	13.8	14.3			
	5% DB	14.9	20.4	23.9	25.5	28.7	27.8	24.7	26.1	24.2	20.8	14.1	13.5			
(36) Mean Daily Temperature Range	MCDBR	13.8	17.1	16.8	14.9	13.9	11.1	10.3	11.8	13.2	13.4	11.0	12.6			
	MCWBR	14.6	19.6	21.2	23.1	25.8	22.6	20.1	22.3	22.4	18.1	13.2	13.3			
	MCWBR	13.9	16.7	15.0	14.5	12.9	9.8	9.8	11.1	12.5	12.0	10.9	12.5			
	MCWBR	13.9	16.7	15.0	14.5	12.9	9.8	9.8	11.1	12.5	12.0	10.9	12.5			
(40) Clear-Sky Solar Irradiance	taub	0.282	0.312	0.341	0.377	0.428	0.428	0.435	0.397	0.350	0.338	0.305	0.280			
	taud	2.015	2.003	2.023	2.190	2.116	2.221	2.226	2.316	2.407	2.278	2.115	1.980			
	Ebn at Noon	189	231	256	266	258	260	255	258	257	225	181	156			
	Edh at Noon	22	32	40	40	46	42	41	35	28	25	20	18			
(44) All-Sky Solar Radiation	RadAvg	285	611	1102	1476	1666	1867	1698	1404	940	521	266	185			
	RadStd	22	29	55	109	152	133	144	89	102	51	28	22			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page